

Offshore e-Methanol Conceptual Engineering Study

Cantabrian Sea

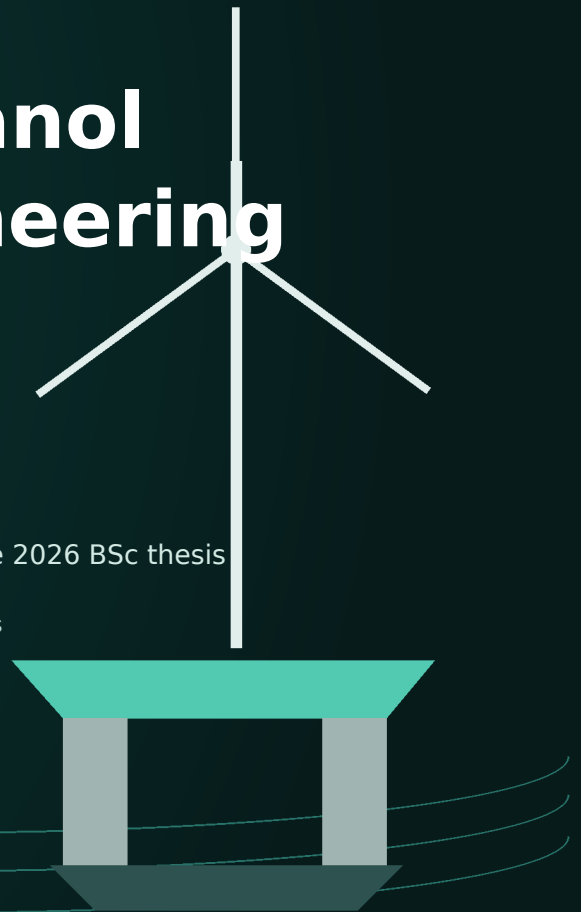
Professional, auditable and reproducible evolution of the 2026 BSc thesis

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Academic origin: Universidad de Cantabria

Original thesis supervisor: Prof. Dr. María Victoria Biezma Moraleda

Professional Edition EN · v1.2 · 13 September 2026



15 MW

naval unit

150 MW

commercial hub

53.2 kt/y

base e-methanol

639 GWh/y

net energy



Interactive model

DOCUMENT STATUS

Edition note, scope and responsibility

This document is an independent professional edition developed by the author from the 2026 BSc thesis. The original thesis remains the deposited academic document; this edition does not replace it, alter its assessment, or imply review, approval or endorsement by Universidad de Cantabria or the original thesis supervisor. MethaNor shows how an academic concept can be re-audited using physical conservation, source traceability, reproducible software and conceptual-engineering discipline.

Limit of use

MethaNor operates at concept / pre-FEED level. It does not replace RAO hydrodynamics, CFD, global or fatigue FEA, coupled mooring design, HAZID/HAZOP studies, hazardous-area classification, environmental permitting, a complete official GIS assessment, vendor quotations or financial due diligence.

Engineering unit 15 MW 1 IEA reference turbine	Commercial architecture 150 MW 10 x 15 MW	Base production 53,184 t/y derived result	Base NPV -303 €m scenario, 8% WACC
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Evidence convention

Every material figure is interpreted under one of four evidence classes. This labelling prevents official data from being mixed with assumptions or calculated results.

Class	Meaning	Use in this study
PRIMARY / OFFICIAL	Technical, government, institutional or manufacturer source.	IEA 15 MW turbine, DOE PEM, ERA5, EMODnet, POEM, Puertos del Estado, IRENA, IMO, FuelEU.
ACADEMIC BASELINE	Value inherited from the thesis and re-audited before reuse.	Coordinates, 8.63 m/s mean wind, k = 2.2, 24,000 t, 22 m draft, 24.5 m air gap, GMeff = 3.13 m.
SCENARIO ASSUMPTION	Editable business, project or planning input.	CAPEX, OPEX, e-methanol price, WACC, CAPEX support, availability and consumptions not fixed by primary evidence.
DERIVED RESULT	Value recalculated from visible equations.	Annual energy, production, mass balance, NPV, LCOeM, payback and break-even thresholds.

NAVIGATION

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MethaNor Live · GitHub repository · Author LinkedIn. External primary-source links are included at the end of the document.

00 • EXECUTIVE SUMMARY

A valid concept, re-audited and rescaled through energy conservation

MethaNor studies an offshore architecture that converts Cantabrian Sea wind resource into e-methanol through electrolysis, CO₂ capture, chemical synthesis, storage and marine logistics. The main contribution of this edition is not a “closed” or construction-ready platform, but the transformation of a multidisciplinary thesis into a traceable decision model: every number has provenance, every equation is exposed, every economic assumption is editable and every evidence gap becomes an engineering gate.

The original thesis defined a 15 MW semisubmersible platform and reported annual e-methanol production of 51,058 t/y. Rebuilding the system as executable software exposed a central inconsistency: even at 100% capacity factor, a single 15 MW source has an absolute theoretical ceiling of 131.4 GWh/y, which is insufficient for that output when hydrogen stoichiometry alone requires about 9.82 electrical MWh per tonne of methanol at the assumed PEM intensity of 52 kWh/kg-H₂. MethaNor resolves the contradiction by clearly separating the 15 MW naval engineering unit from a 150 MW commercial hub.

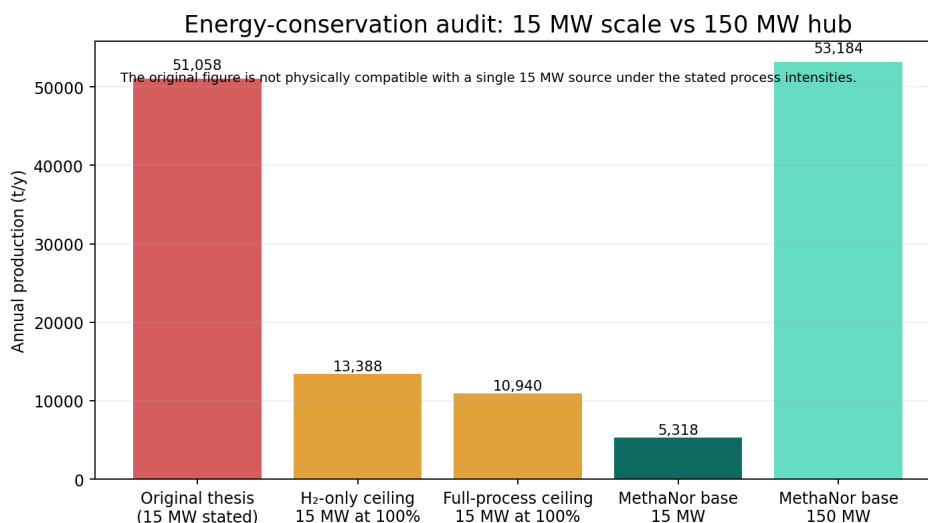


Figure 1. Energy-conservation audit. The academic figure of 51,058 t/y is not compatible with a single 15 MW source under the stated process intensities; the order of magnitude becomes physically coherent when scaled to a 150 MW hub.

Gross CF 51,7% Weibull + screening curve	Availability 94% base assumption	Net energy 638.8 GWh/y 150 MW hub	Total intensity 12.01 MWh/t includes 6% BoP
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With these values, the base model produces 53,184 t/y of e-methanol, stoichiometrically consumes 10,039 t/y of H₂ and requires 73,048 t/y of CO₂. Applying the same formulation to a single 15 MW unit yields about 5,318 t/y, consistent with the energy constraint and preserving the naval concept without overstating its commercial output.

Base economic conclusion

With gross CAPEX of €1,000m, a 40% CAPEX-support scenario, OPEX of €33m/y, an e-methanol price of €1,200/t, 8% WACC, 0.5% annual production degradation and terminal decommissioning equal to 3% of gross CAPEX, the model returns NPV ≈ -€303m, LCOeM ≈ €1,755/t and simple payback ≈ 21.7 years. A conventional IRR is not reported because the base-case cash-flow pattern does not yield a decision-useful internal root within the calculation interval. The result must be read as conceptual screening, not as a bankable valuation.

From a 2026 BSc thesis to an auditable conceptual engineering study

The academic source document, titled “Multidisciplinary Conceptual Design and Techno-Economic Feasibility of a 15 MW P2X Semisubmersible Platform for e-Methanol Synthesis in the Cantabrian Sea”, was developed for the BSc in Marine Engineering at Universidad de Cantabria under the supervision of Prof. Dr. María Victoria Biezma Moraleda. Its original scope covered site selection, wind resource, P2X, dynamic modelling, naval architecture, stability, safety, CAPEX/OPEX and financial scenarios.

This edition preserves what remains useful as an engineering hypothesis, while removing or downgrading any claim that exceeds the available evidence. That discipline matters because the original thesis contains statements that are too conclusive for the actual maturity of the model, including strict carbon-neutrality claims, regulatory compliance, ATEX declassification, quantified aerodynamic reductions and final technical-operational viability without a complete LCA, class-grade FEA/CFD, HAZOP or complete official GIS intersections.

Topic	2026 thesis	MethaNor - professional edition
Energy scale	15 MW and 51,058 t/y are linked in the narrative.	15 MW = naval unit; 150 MW = commercial hub. Production is always energy-limited.
Production	P50 ≈ 51,058 t/y; commercial P90 ≈ 40,000 t/y.	53,184 t/y base for 150 MW; 5,318 t/y for 15 MW under the same assumptions.
Economics	NPV/IRR and 40% support mixed across different CAPEX values and bases.	Single reproducible DCF tied to the current physical result; support and WACC are scenarios, not entitlements.
Site	Coordinates and suitability conclusions based on academic maps.	Coordinates preserved; permitting, AIS, protected areas and bathymetry remain open until official GIS verification.
Marine rules	The text refers to “meeting” DNV criteria.	DNV is used as a design reference; class approval is not claimed.
Carbon	Strict carbon neutrality is stated.	Well-to-wake / LCA assessment and RFNBO compliance are required before claiming GHG intensity.
HSE	Zoning/ventilation are presented as validated.	Safety concept only; HAZID/HAZOP and IEC 60079 are future gates.

Correction principle

The project becomes more valuable when it states its limits. A figure is removed if it cannot be defended; retained as an academic baseline when it provides continuity; updated when a stronger primary source exists; and recalculated when it derives from a verifiable equation.

02 • SYSTEM ARCHITECTURE

From wind to molecule: physical and logical system boundary

The base architecture is modular. Ten 15 MW reference wind units form a 150 MW hub feeding a centralized or semi-distributed P2X train. The concept preserves a 15 MW semisubmersible as the naval integration unit while the hub represents commercial scale. This separation prevents the mass, power and chemical capacity of the entire complex from being assigned to one platform.

Annual mass and energy balance - base case

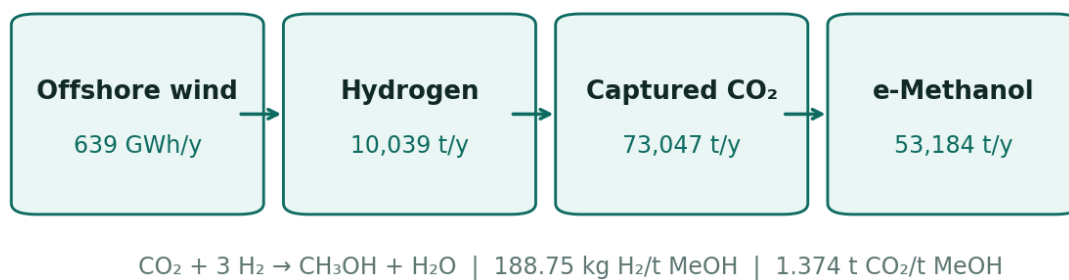


Figure 2. Annual mass and energy balance for the base case. H₂ and CO₂ flows are derived stoichiometrically from e-methanol production.

2.1 Model limits

The calculation combines a stationary Weibull wind resource, a transparent power curve, availability, equivalent electrical process intensities, stoichiometric mass balance and a 25-year cash-flow model. It does not include OpenFAST aeroelastic dynamics, inter-turbine wake losses, optimized hourly dispatch, detailed electrochemical storage, reactor kinetics, multistage compression, rigorous distillation, steam-network design, equipment-specific physical degradation or weather-correlated unavailability. These omissions are stated explicitly to avoid false precision.

2.2 Governing equation

Production is calculated from annual available energy and total process intensity rather than setting a target output first and solving backwards for the required energy.

ANNUAL WIND ENERGY

$$E_{\text{wind}} = N \cdot P_{\text{rated}} \cdot 8.760 \cdot CF \cdot A$$

N: number of turbines · CF: gross capacity factor · A: availability

EQUIVALENT ELECTRICAL INTENSITY

$$I_{\text{total}} = (I_{\text{PEM}} + I_{\text{DAC}} + I_{\text{MeOH}} + I_{\text{water}}) / (1 - f_{\text{BoP}})$$

All terms are expressed in MWh per tonne of e-methanol.

ENERGY-LIMITED PRODUCTION

$$m_{\text{MeOH}} = E_{\text{wind}} / I_{\text{total}}$$

Production is not fixed a priori; it follows from the energy actually available and the total specific consumption.

Symbol	Interpretation	Unit
N	Number of reference turbines	-
P_{rated}	Rated power per turbine	MW
CF	Gross screening capacity factor	-
A	Technical availability	-
I_{total}	Total equivalent electrical intensity	MWh/t MeOH
m_{MeOH}	Annual e-methanol production	t/y

Key result

Energy conservation acts as a binding physical constraint. Any change in wind resource, PEM efficiency, DAC, synthesis, availability or balance of plant automatically changes production and therefore the economic metrics.

03 · SITE AND METOCEAN

Screen the Cantabrian Sea before sizing steel

The thesis located the platform geometric centre at 43°35'35" N, 3°51'00" W (approx. 43.593° N, 3.850° W), with an academic water depth of 100 m and a mean wind resource of 8.63 m/s at 150 m. The original work used the vicinity of Puertos del Estado SIMAR node 3136040 as an oceanographic reference and adopted $H_{s,50} = 11$ m as a conceptual survival value. In MethaNor these figures remain an academic baseline, not a certified site-characterisation campaign.

Variable	Current baseline	Class	Required professional upgrade
Coordinates	43.593° N; -3.850° W	Academic	Check spatial conflicts against POEM, protected areas, cables, fisheries, defence constraints and AIS.
Water depth	100 m	Academic	Sample the EMODnet DTM and confirm vertical datum / resolution.
Mean wind at hub height	8.63 m/s	Academic	Multi-year hourly ERA5 plus consistent height extrapolation, or a validated mesoscale resource.
Weibull k	2,2	Academic	Replace with an hourly time series; retain Weibull only for conceptual screening.
$H_{s,50}$	11.0 m	Academic	Wave extremes and peak period from Puertos del Estado / hindcast with dedicated statistical analysis.
Port distance	~32 km to Santander in the base UI	Derived / screening	Define O&M; and fabrication ports with actual draft, crane and laydown-area constraints.

3.1 The missing GIS layer

A satellite basemap provides visual context, not permitting evidence. The web application links ERA5, EMODnet, POEM and Google Maps and avoids claims such as “too much shipping”, “protected area” or “impossible site” unless the corresponding GIS/AIS layer is available. For pre-FEED, the site gate should integrate at least AIS traffic density, POEM polygons, Natura 2000 and other protected areas, bathymetry, cables and pipelines, fisheries, military constraints, aviation routes, ports and geotechnical data.

Site-screening discipline

The app score (wind 35%, bathymetry 20%, metocean 15%, logistics 15%, spatial planning 15%) is a portfolio-prioritisation index. It is not an engineering acceptance criterion or a permitting opinion.

04 • WIND RESOURCE

Open reference curve, visible assumptions and an upgrade path to hourly data

MethaNor uses the IEA Wind TCP 15 MW reference turbine, a public tool designed for research and offshore design comparison. The reference geometry includes a 240 m rotor, 150 m hub height, 3 m/s cut-in, 10.59 m/s rated wind speed and 25 m/s cut-out. Between cut-in and rated speed, a transparent cubic power interpolation is used: adequate for conceptual screening, but not equivalent to an aeroelastic controller or OpenFAST.

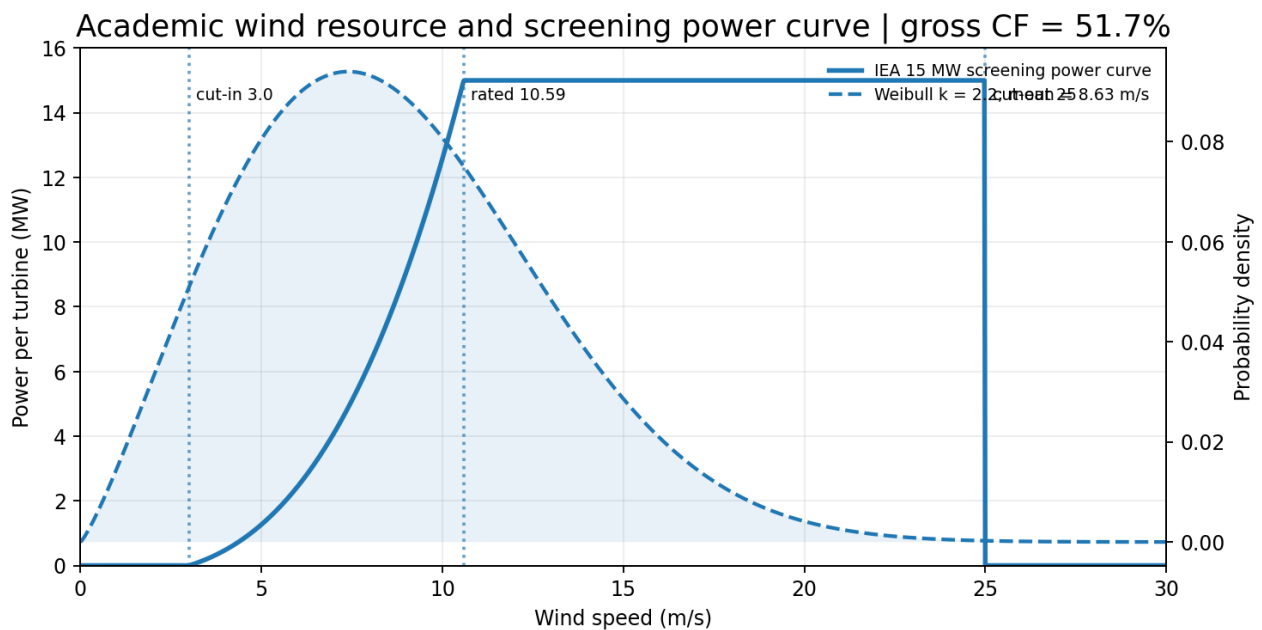


Figure 3. Academic Weibull distribution and screening power curve. The calculated gross capacity factor is ~51.7%; with 94% availability, net hub energy is ~638.8 GWh/y.

4.1 Data upgrade

The professional version should migrate from a static Weibull distribution to a multi-year hourly time series. ERA5 provides hourly reanalysis from 1940 and enables reproducible reconstruction of wind, waves and interannual variability. A pre-FEED pipeline should download the relevant point or grid-cell series, apply quality control, extrapolate consistently to hub height, propagate losses, calculate P50/P75/P90 and preserve each transformation with data version and hash. Resource uncertainty must propagate into the financial model rather than remaining isolated in the metocean chapter.

What is not claimed

8.63 m/s is not presented as an in-situ measurement. The 51.7% capacity factor is not treated as “bankable”. Both are screening results that must be replaced by a project-specific wind-resource assessment.

Chemical and energy balance of the e-methanol train

The base reaction is $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$. Molar masses imply approximately 188.75 kg H_2 and 1.374 t CO_2 per tonne of methanol. MethaNor derives these factors from stoichiometry to avoid inserting untraceable production yields.

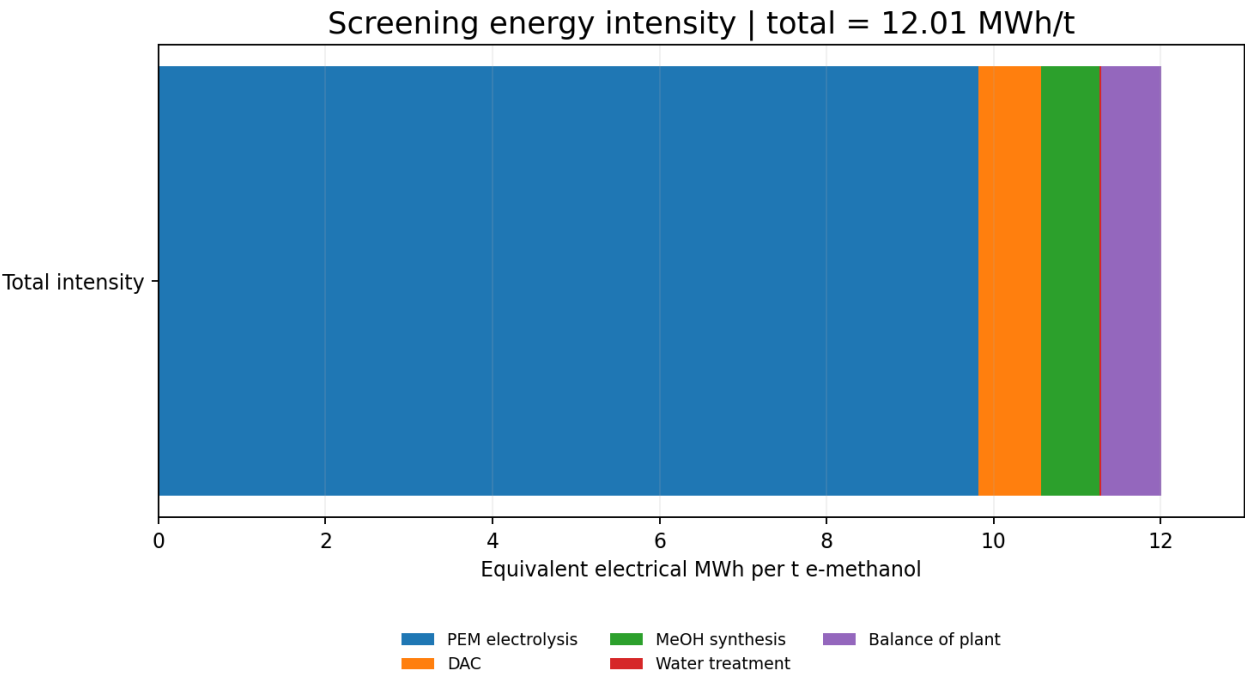


Figure 4. Equivalent electrical intensities for the base case. Electrolysis dominates consumption; DAC, synthesis, water treatment and balance of plant bring the total to ~12.01 MWh/t e-methanol.

Block	Base value	Evidence / status	Comment
PEM	52,0 kWh/kg H ₂	Scenario close to DOE reference	DOE sets the 2026 PEM system target at 51 kWh/kg H ₂ ; 52 is retained as a conservative screening value.
DAC	0.55 MWh/t CO ₂	Scenario; IEA as context	Total DAC energy depends on technology and heat supply; it is not treated as vendor data.
MeOH synthesis	0.70 MWh/t MeOH	Scenario / literature	Simplified electrical equivalent; it does not replace rigorous process simulation.
Water	0.02 MWh/t MeOH	Scenario	Treatment order of magnitude; requires a full water and brine balance.
BoP	6% of gross demand	Scenario	Pumps, compression, controls, losses and auxiliaries aggregated.

5.1 Annual result

e-Methanol 53,184 t/y energy-limited	H ₂ 10,039 t/y stoichiometric	CO ₂ 73,048 t/y stoichiometric	O ₂ coproduct 79,667 t/y not monetised in base case
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Water generated by the reaction itself is approximately 29,902 t/y. Oxygen from electrolysis is retained as a technical coproduct but is not monetised in the base financial case to avoid inserting uncontracted revenue. The same applies to carbon credits, guarantees of origin and other environmental attributes: they should enter cash flow only under verifiable rules and contracts.

5.2 Carbon: from “neutral” to traceable

The thesis described e-methanol as strictly carbon neutral. That wording is removed. GHG intensity depends on electricity origin, CO₂ origin and capture, thermal demand, construction, logistics and end use. IMO is moving toward life-cycle assessment and FuelEU Maritime regulates the GHG intensity of energy used on board. In the EU, RFNBO classification also imposes rules on renewable electricity, additionality and temporal/geographical correlation. MethaNor therefore refers to GHG-reduction potential, never automatic neutrality.

Select an architecture without inventing quotations

The thesis fixed specific solutions such as Siemens Silyzer 300, Climeworks and Topsoe. A professional edition should separate technology from vendor. The MethaNor app includes a public-data candidate explorer for electrolysis, methanol synthesis, DAC and water treatment; it scores concept fit using visible criteria, but does not claim a universal “best vendor” or fill in unpublished commercial fields.

Category	Indicative shortlist	Decision criteria
Electrolysis	Nel MC Series, ITM POSEIDON, Siemens Energy Elyzer P-300, thyssenkrupp nucera scalum, Plug Power, John Cockerill, Sunfire, Enapter, Topsoe y otros.	Module power, dynamic response, pressure, purity, maturity, integration and quality of public evidence.
MeOH synthesis	Commercial catalytic routes and licensable packages.	Pressure/temperature, turndown, heat integration, maturity, minimum scale and guarantees.
DAC / CO ₂	Solid/liquid DAC and biogenic/captured CO ₂ alternatives.	Thermal and electrical demand, CO ₂ quality, modularity, marine sensitivity and carbon chain.
Water	Seawater RO + polishing for electrolysis.	Consumption, quality, recovery, brine, redundancy and marinisation.

Technical procurement rule

A “public manufacturer page” is not a quotation. If price, availability, guaranteed performance or EPC scope is not defensibly published, the document must state “quotation required”. Comparative literature references are kept separate from vendor pricing.

6.1 Why PEM remains the baseline

PEM remains the baseline because of its dynamic response and conceptual compatibility with variable renewable power. However, a pre-FEED study should compare PEM, pressurised alkaline and SOEC against the hourly energy profile, intermediate storage, target pressure and thermal strategy. The decision must not be based on kWh/kg alone: installed CAPEX, degradation, stack replacement, water treatment, compression and availability may reverse the ranking.

The interactive comparison, manufacturer sources and technical links are retained in the MethaNor Live explorer.

07 • FLOATING PLATFORM

Preserve the naval concept without presenting it as class-approved design

The semisubmersible is retained as the visual and engineering core of the project: four columns, pontoons/cross-members, elevated process deck, central wind-turbine support, storage, helideck and catenary mooring. Values of 24,000 t displacement, 22 m draft, 24.5 m air gap and $GMeff = 3.13$ m are treated as an academic reference. They support design continuity but do not constitute approval of stability, structure or seakeeping.

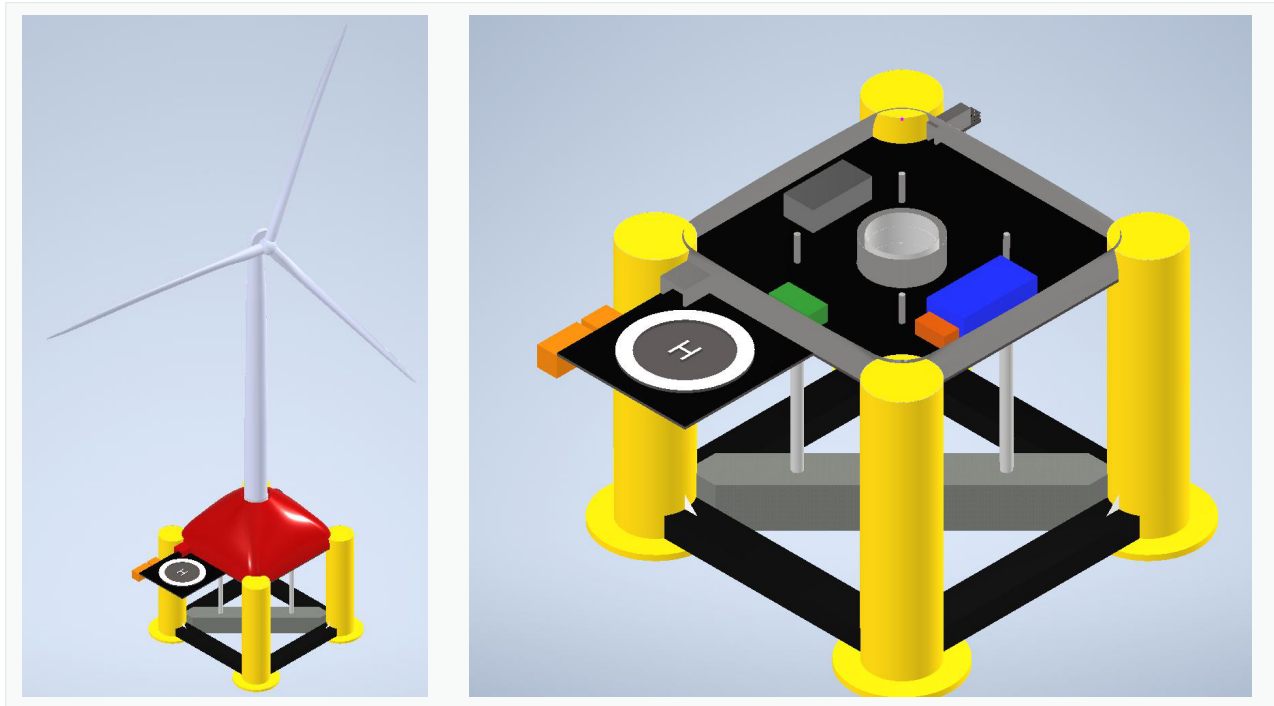


Figure 5. Original academic CAD (2026). Left: overall unit view. Right: deck integration. Images retain their original proportions and are used as conceptual references, not as an approved General Arrangement.

7.1 Decisions that survive the re-audit

- Semisubmersible: consistent with water depths of roughly 100 m and the need for a large integration deck, although geometry must be optimised through coupled hydrodynamic and structural analysis.
- High air gap: retained as a robustness objective against green water/slamming, but it must be recalculated using wave extremes, 6-DOF motions and class criteria.
- Decoupled load paths: retained as an architectural intent to limit transmission of wind-turbine loads into process modules; requires global FEA and fatigue analysis.
- Protective superstructure: retained as a marination concept, while quantified aerodynamic reduction, ACH and ATEX-declassification claims not supported by CFD/ventilation/IEC 60079 are removed from the main conclusions.
- Catenary mooring: retained as a starting point; final design requires soil data, lines, anchors, offsets, coupled loads and dynamic verification.

07B · NAVAL INTEGRATION

General arrangement: from a 3D idea to a verifiable architecture

To preserve the depth of the original thesis, this edition retains selected original integration views. Their purpose is not decorative: they trace how the process plant, access routes, deck and protective envelope constrain mass, centres, safety, maintenance and marine operations.

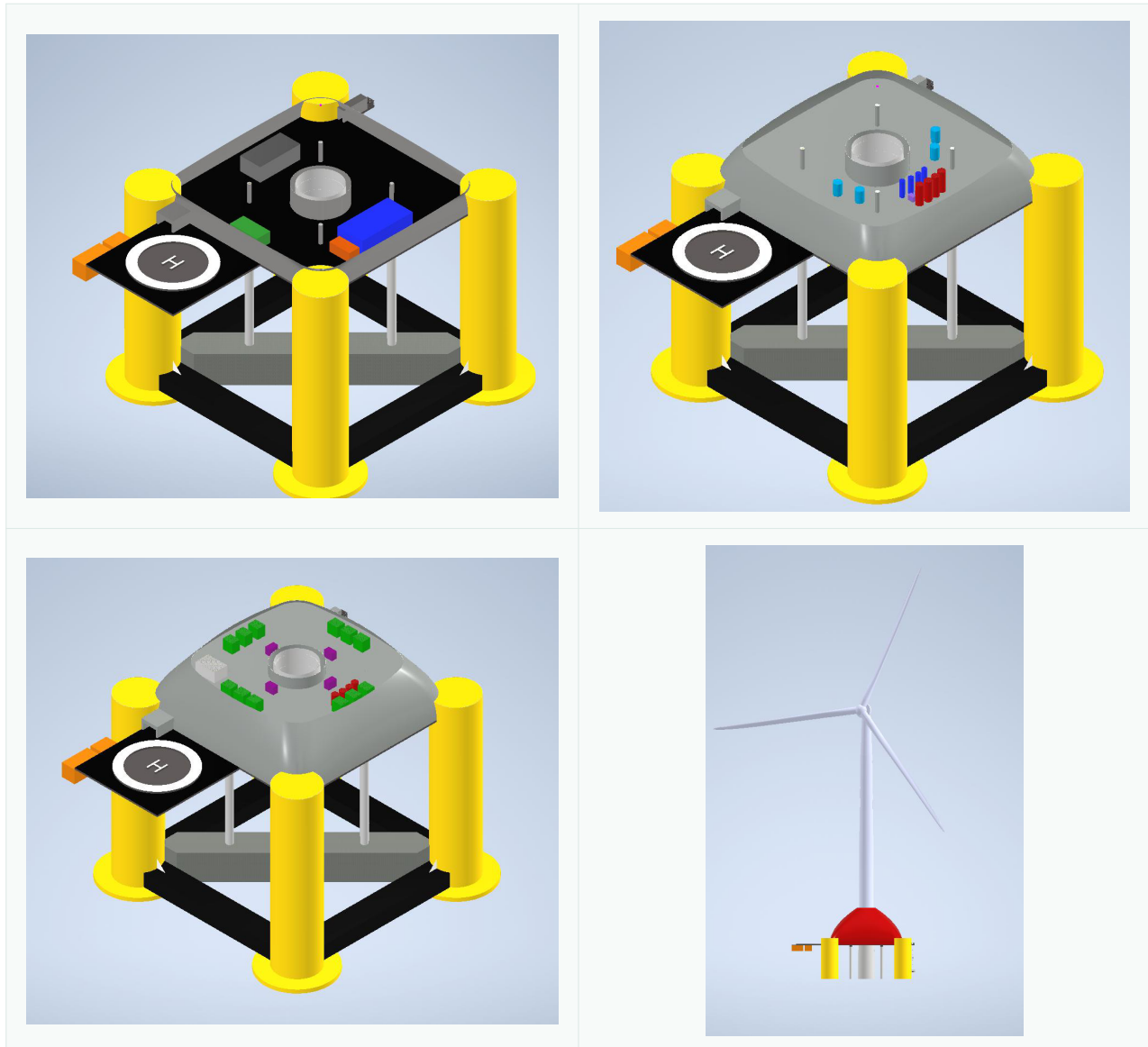


Figure 6. Selected original thesis views: process-level integration and air-gap reference. The professional edition retains them as evidence of geometric development while explicitly identifying which class verifications remain open.

Engineering interpretation

Concept depth is demonstrated by connecting geometry to requirements: escape routes, inventory segregation, equipment maintenance, global loads, centre of gravity, ventilation, marine access and space reserve. An image alone closes none of these requirements.

08 • MASS, SPACE AND HYDROSTATICS

Connect every equipment choice to naval reality

One strength of the thesis is that it did not treat the chemical plant as a weightless diagram: it built a mass and vertical-moment register closing at 24,000 t and 505,757 t·m, leading to $KG \approx 21.07$ m. MethaNor preserves this register to maintain spatial discipline. The limitation is equally important: process and superstructure masses are academic values and must be replaced by vendor data, MTO and structural estimates as the design matures.

Item	Mass (t)	VCG (m)	Moment (t·m)
Naval substructure	4.500	15,0	67.500
Fixed ballast	6.000	2,0	12.000
Active ballast	7.500	5,0	37.500
Stored MeOH (7 days)	1.500	12,0	18.000
Suspended mooring	500	10,0	5.000
Central column	450	39,0	17.550
IEA 15 MW tower	860	109,0	93.740
RNA	1.187	150,0	178.050
Electrolysis	408	46,5	18.972
RO + auxiliaries	150	46,5	6.975
MeOH synthesis	350	52,0	18.200
DAC	120	57,0	6.840
Superstructure	300	58,0	17.400
Helideck + helicopter	155	46,0	7.130
Life-saving equipment	20	45,0	900
TOTAL	24.000		505,757

Correct interpretation

Mass closure is valuable as bookkeeping, but a “balanced weight” does not validate stability or strength. FEED requires lightweight/deadweight control, discipline-specific uncertainty, MTO, actual centres, free-surface effects, damage cases, intact-stability assessment and class criteria.

8.1 Hydrodynamic gate

The next phase should replace simplified checks with a coupled hydrodynamic model delivering RAOs, motions, accelerations, dynamic air gap, green water, line loads, wind/wave/current response and operational weather windows. Global FEA, connection detailing and fatigue can then be closed. Until then, “ $G_{Meff} = 3.13$ m” remains an academic result, not a stability certificate.

Turn uncertainties into gates before they become sunk cost

The system combines hydrogen, methanol, oxygen, high-power electrical equipment and remote marine operation. The thesis developed concepts for ATEX zoning, ESD, gas detection and corrosion protection; the professional edition retains them as design intentions and avoids presenting them as validated. The next work stage should be structured around HAZID/HAZOP, isolation philosophy, blowdown/vent, F&G, SIS, consequence analysis, escape/evacuation and hazardous-area classification.

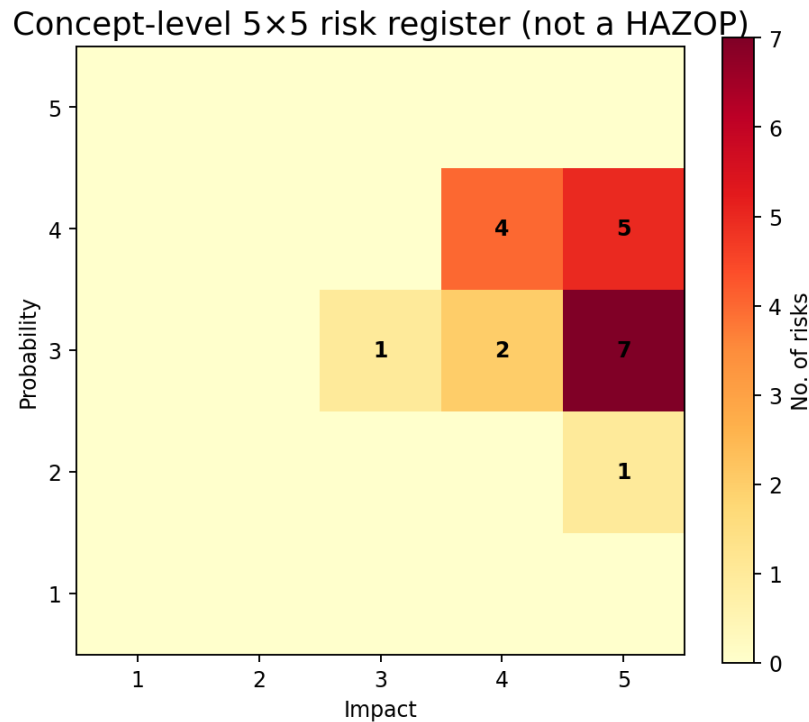


Figure 7. Concept-level 5x5 MethaNor risk matrix. It is a prioritisation tool, not a HAZOP.

Risk	Priority	Mitigation gate
Metoccean basis not closed	Critical	Official time series, extremes, campaign and design criteria.
Insufficient bathymetry / geotechnics	Critical	DTM + geophysics/geotechnics + anchor/mooring selection.
Navigation / use conflict	Critical	AIS and official GIS layers; do not decide from an illustrative map.
H ₂ / hazardous areas	Critical	HAZID/HAZOP, IEC 60079, ventilation, F&G; and ESD.
MeOH fire/toxicity	High	Inventory, containment, drainage, foam/water, detection and segregation.
CAPEX and supply chain	High	Class 4/3 estimate, budgetary quotations, contingency and escalation.
e-Methanol price/offtake	Critical	Offtake agreement, certification, sensitivity and support structure.
Weather-limited O&M; access	High	Weather windows, SOV/CTV/helicopter, spares and remote diagnostics.
Marine corrosion	High	Modern ISO 12944 specification, materials selection and cathodic protection.

10 • BUILD AND OPERATIONS

Translate technology into a project people can actually build, integrate and operate

MethaNor includes a conceptual 60-month schedule because an offshore concept cannot be assessed only through thermodynamic efficiency. The critical path depends on site campaigns, permits, class basis, long-lead equipment, hull fabrication, module integration, moorings, final connection and commissioning. Durations are conceptual assumptions and must be replaced by a project master schedule and yard/vendor data.

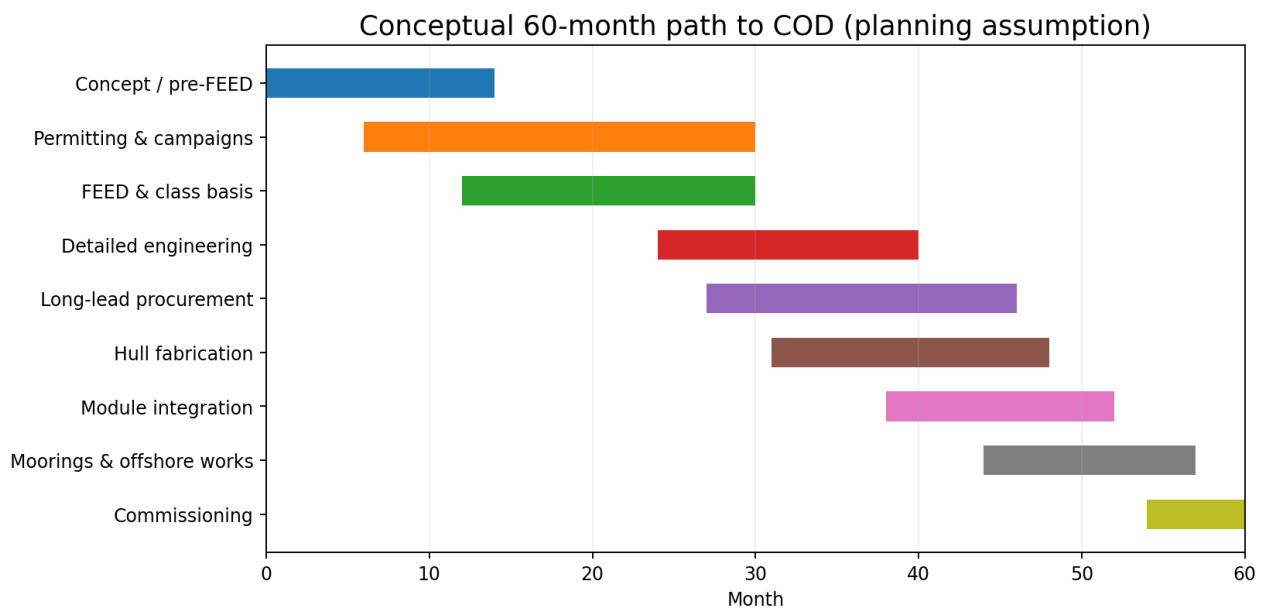


Figure 8. Conceptual 60-month path to COD. It is a planning hypothesis used to structure gates, not a contractual programme.

Phase	Indicative staffing	Interpretation
Concept / pre-FEED	18 FTE	Multidisciplinary engineering, data, HSE, cost and permitting.
FEED and permitting	42 FTE	Higher workload across process, marine, electrical/control and environmental disciplines.
Detailed engineering	78 FTE	IFC packages, vendor data, coordination and interfaces.
Fabrication / integration	115 FTE	Construction peak; does not include the yard's full total workforce.
Commissioning	64 FTE	System completion, testing, HSE and commissioning.
Steady operations	15 FTE	Remote-first operating model with campaigns; pending RAM and weather-window analysis.

10.1 O&M; philosophy

Remote operation reduces offshore exposure but does not eliminate people or logistics. The 94% base-case availability must be decomposed into reliability, maintainability and weather accessibility. A professional RAM model should translate failure rates, MTTR, spares and sea-state constraints into annual availability, campaign cost and lost production. Only then can 15 FTE, a campaign frequency or an SOV/CTV/helicopter strategy be justified.

11 • TECHNO-ECONOMICS

Make every financial conclusion depend on visible assumptions

The economics in this edition are recalculated from the current physical result. This avoids the issue in the original thesis where different tables mixed 40,000 t/y, 51,058 t/y, gross/net CAPEX and support assumptions on partially inconsistent bases. MethaNor defines one base case, applies annual degradation and a decommissioning provision, and calculates NPV, LCOeM, payback and break-even conditions using the same equations.

Base input	Value	Classification
Gross CAPEX	€1,000m	Scenario assumption, high uncertainty
CAPEX support	40%	Policy scenario; no entitlement
Net CAPEX	€600m	Derived
Annual OPEX	€33m/y	Scenario assumption
e-Methanol price	€1,200/t	Market / offtake scenario
WACC	8%	Financing assumption
Life	25 years	Assumption
Production degradation	0.5%/y	Assumption
Decommissioning	3% of gross CAPEX	Assumption

NPV -303 €m @ 8% WACC	LCOeM €1,755/t levelised cost	Payback 21.7 years simple, undiscounted	Year-1 margin €30.8m before financing
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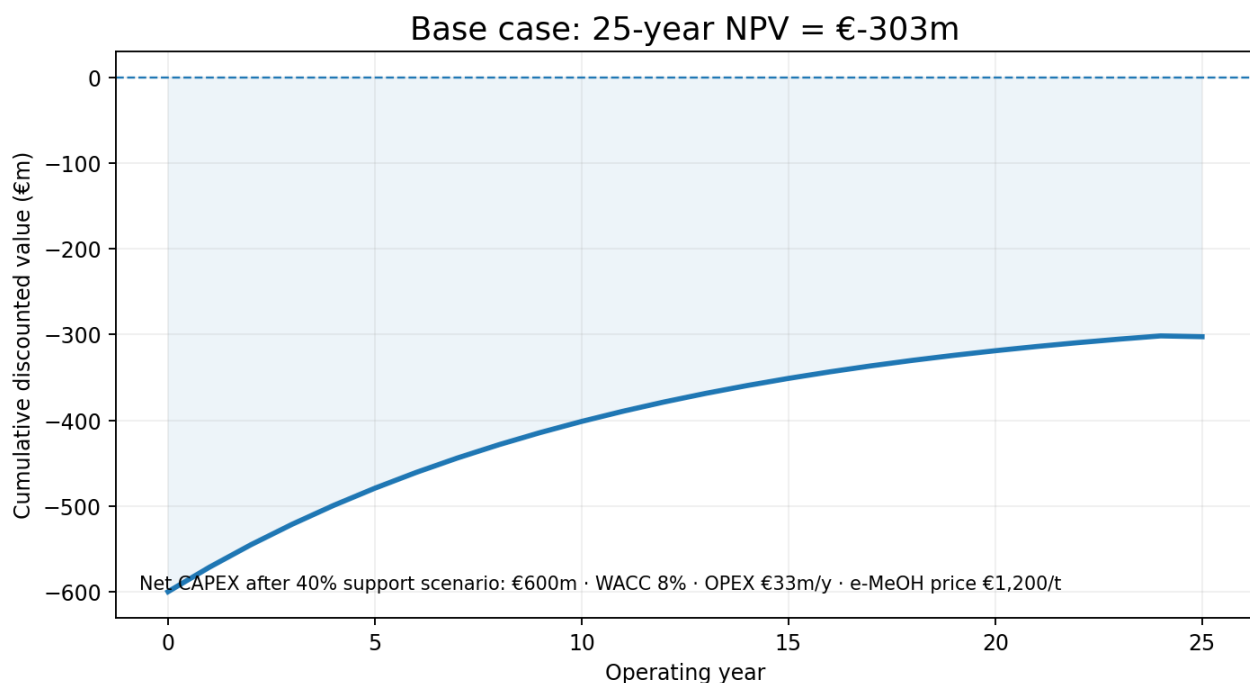


Figure 9. Cumulative discounted value of the base case. The project does not reach positive NPV under the reference assumptions.

11.1 Cost structure and sensitivity

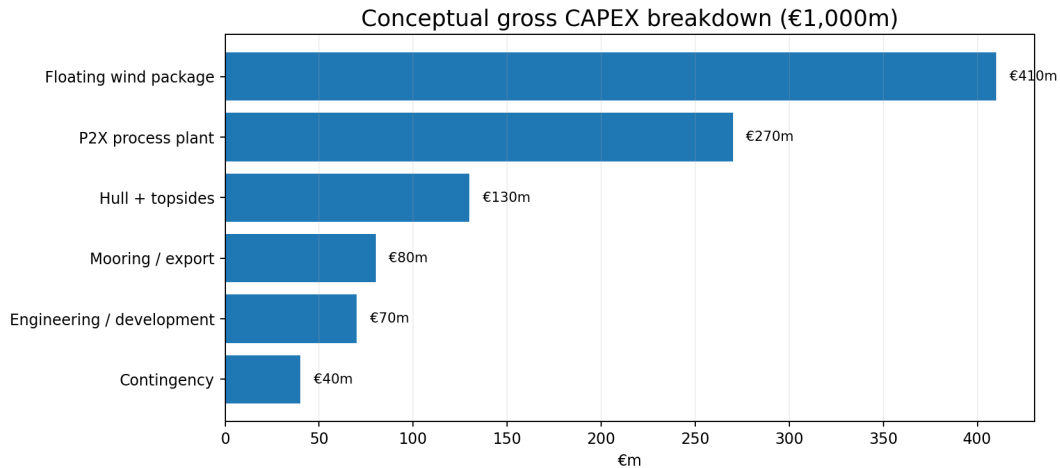


Figure 10. Conceptual gross CAPEX breakdown. Values are not vendor quotations.

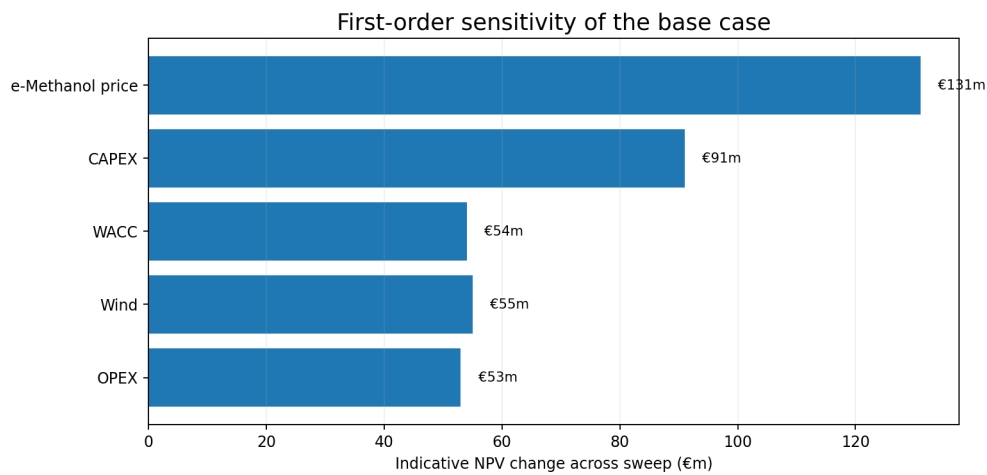


Figure 11. Indicative sensitivity. e-Methanol price and CAPEX dominate NPV variability across the model sweep.

Variable solved for NPV = 0	Base result	Interpretation
e-Methanol price	~€1,755/t	Break-even price under all other assumptions.
Maximum gross CAPEX	~€499m	With 40% support, 8% WACC and €1,200/t price.
Minimum CAPEX support	~70.3%	Must not be interpreted as available support; it is only a mathematical condition.
Minimum mean wind	Not reachable in tested range	Resource alone does not offset the cost structure.
Maximum WACC	~0.67%	Shows that cheaper capital alone is not realistically sufficient without changing other variables.

Public-policy correction

The thesis treated 40% Innovation Fund support and CfD support as near-deterministic enablers. This edition converts both into scenarios. The Innovation Fund selects projects through competitive calls and auctions with specific rules; there is no generic entitlement to 40% of CAPEX. A serious project must model support as a variable and verify the terms of the applicable call.

12 · REGULATORY FRAMEWORK AND DECARBONISATION

The molecule is only “green” if the full chain proves it

The 2023 IMO GHG Strategy sets the ambition of reaching net-zero GHG emissions from international shipping around 2050, with indicative GHG-reduction checkpoints for 2030 and 2040 and an uptake objective for zero- or near-zero-emission technologies/fuels. FuelEU Maritime progressively limits the annual average GHG intensity of energy used on board. This creates regulatory demand for e-fuels, but does not by itself guarantee MethaNor’s commercial viability.

Block	Implication for MethaNor
IMO 2023 GHG Strategy	The product must demonstrate a real GHG-reduction contribution; assessment shifts toward life cycle.
FuelEU Maritime 2023/1805	Fuel well-to-wake intensity matters; commercial value depends on certification and compliance.
RFNBO - Reg. Delegado (UE) 2023/1184	Renewable electricity may be subject to additionality and temporal/geographical correlation. A dedicated offshore system may simplify traceability but requires robust legal configuration and metering.
Spanish POEM	The site must be intersected with official layers and use-compatibility constraints.
DNV / IEC	The platform and plant require a design basis, class involvement, functional safety and hazardous-area engineering before FEED.

12.1 Minimum carbon-accounting basis

Before publishing a “footprint reduction” figure, the system boundary, electricity mix/attribution, turbine and floater manufacturing, replacements, marine transport, captured CO₂, leakage/venting, methanol distribution and end use must be defined. The result should be expressed in gCO₂e/MJ or another regulatory-compatible unit, with scenario and source. The academic “>91%” figure is therefore removed from the main claim until a reproducible LCA is available.

13 · MATURITY ROADMAP

How to turn MethaNor into a defensible pre-FEED package or European proposal

MethaNor already functions as a decision architecture and digital demonstrator. To move it toward a European project, the priority is not more screens but closing evidence. The quality step occurs when each “required data” item becomes a work package, deliverable, owner, acceptance criterion and investment gate.

MethaNor technical maturity roadmap

From auditable concept to demonstrable pre-FEED package

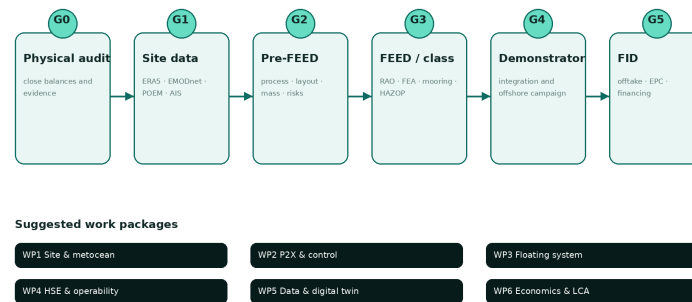


Figure 12. Proposed maturity roadmap. The sequence clearly separates conceptual screening, pre-FEED, FEED/class, demonstration and FID.

WP	Objective	Minimum deliverables
WP1 - Site	Close resource, metocean, geotechnical and spatial constraints.	Versioned dataset; P50/P90; extremes; GIS constraints; site shortlist.
WP2 - P2X and control	Define balances and operating dynamics under variable wind.	PFD/P&ID; basis; process simulation; storage sizing; operating philosophy.
WP3 - Floating system	Close semisubmersible, masses, hydrodynamics and mooring.	General arrangement; weight report; RAOs; mooring; air gap; global loads; FEA basis.
WP4 - HSE and operability	Design safety from the architecture.	HAZID/HAZOP, hazardous areas, F&G/ESD, escape and marine operations.
WP5 - Digital twin	Connect data, models and reproducible scenarios.	ERA5/EMODnet flows; versioned engine; uncertainty; CI/tests.
WP6 - Economics and LCA	Move from assumptions to Class 4/3 estimate and regulatory framing.	Budgetary quotations; CAPEX/OPEX; DCF; LCOeM; LCA; financing and offtake scenarios.
WP7 - Demonstration	Prepare a physical campaign or partial integration.	Test plan, TRL evidence, commissioning and performance verification.

Potential fit with EU funding

The Innovation Fund supports demonstration and commercialisation of low-carbon technologies through competitive calls and mechanisms. MethaNor could use its architecture as a project-development basis, but eligibility, support intensity, GHG methodology, minimum maturity and contractual structure must be assessed against the specific call. Support is not treated as an entitlement.

Executable equations instead of results hidden in a PDF

The most important evolution from the thesis is not graphical but methodological. MethaNor implements the physical-financial model independently in Python and JavaScript, generates a deterministic reference case and runs invariant tests. This turns the study into an auditable artefact: when one assumption changes, all related outputs change and the code history preserves the decision trail.

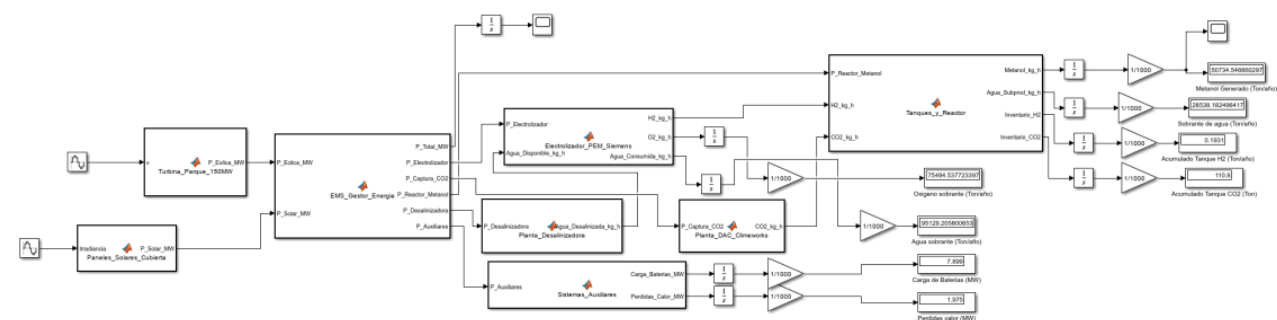


Figure 13. Architecture of the original MATLAB/Simulink dynamic model. The MethaNor edition does not hide this lineage: it separates the academic model from the new screening engine and documents which assumptions are retained, corrected or replaced.

The academic model included wind generation, energy management, electrolysis, desalination, DAC, auxiliaries, the methanol reactor and inventory states. The new software layer is not intended to replace high-fidelity dynamic process simulation; it acts as an independent conservation and scenario engine to test global consistency and prioritise the next engineering work.

Layer	Content	Quality rule
Evidence	ERA5, EMODnet, POEM, Puertos, NREL/IEA, DOE, IRENA, manufacturers.	Source, date, variable, quality and module of use.
Physics engine	Weibull + power curve; energy; stoichiometry; intensities.	Energy and mass conservation; consistent units.
Naval / project model	Mass/space, risks, schedule and O&M.;	Labels “academic” or “scenario” when class/vendor data do not exist.
Economics	25-year DCF, LCOeM, payback and break-even analysis.	Same physical production; no hidden revenue or implicit support.
Interface	EN/ES, map, technology, A/B/C scenarios, evidence.	Do not invent precision; expose sources and evidence gaps.
QA	Pytest, repository validation, CI, JS syntax checks.	Reproducibility before publication.

14.1 How to reproduce

The public repository contains the Python engine, input and reference-case JSON files, documentation, tests and the static web application. The minimum sequence is:

```
python engine/methanor.py python -m pytest -q node --check assets/app.js
```

Repository: <https://github.com/aaronexposito0-creator/methanor-offshore-emethanol-digital-twin>

Application: <https://aaronexposito0-creator.github.io/methanor-offshore-emethanol-digital-twin/>

15 • CONCLUSIONS

What MethaNor demonstrates - and what it must not claim yet

- The system concept remains valuable. Integrating floating wind, H₂, CO₂ and offshore e-methanol is a technically plausible research and development pathway; MethaNor's value lies in connecting disciplines through one auditable balance.
- The 15 MW / 51 kt-y correction was necessary. One unit cannot sustain that production under the stated consumptions. The 150 MW hub restores the commercial scale without discarding the naval concept.
- The base case is not bankable. Under current assumptions, negative NPV and LCOeM above the sale price indicate that the project simultaneously needs lower CAPEX/OPEX, higher revenues, scale, public support and/or more competitive financing.
- The site has not yet been selected professionally. The Cantabria coordinates are an academic reference. A decision requires official GIS, AIS, protected areas, geotechnics, extremes and campaigns.
- The semisubmersible is a conceptual reference, not a class design. It retains a coherent architecture for continued engineering, but needs RAO/CFD/FEA, coupled mooring, a stability book and marine-operations engineering.
- e-Methanol is not automatically declared carbon neutral. Climate benefit must be demonstrated through LCA/well-to-wake assessment and the applicable certification/RFNBO rules.
- Digital traceability is a differentiating strength. MethaNor turns a static academic report into a reproducible and improvable engineering decision-support tool.

Recommended next decision

If MethaNor moved into a real project phase, the next euro should be spent on site evidence, process definition and vendor budgetary data - not on increasing the visual detail of the concept. The pre-FEED objective would be to reduce uncertainty in resource, permitting, masses, CAPEX and safety until a development decision can be taken against measurable criteria.

APPENDIX A

Equations and reproducible reference case

Parameter	Value
Turbines	10 x 15 MW
Mean wind	8.63 m/s
Weibull k	2,2
Availability	94%
Gross capacity factor	51.72%
Net annual energy	638,807 MWh
PEM	52 kWh/kg H ₂
DAC	0.55 MWh/t CO ₂
Synthesis	0.70 MWh/t MeOH
Water	0.02 MWh/t MeOH
BoP	6%
Total intensity	12.011 MWh/t MeOH
e-MeOH production	53,184 t/y
H ₂	10,039 t/y
CO ₂	73,048 t/y

A1 · WEIBULL DISTRIBUTION - SCALE PARAMETER

$$c = \bar{v} / \Gamma(1 + 1/k)$$

$\bar{v}$: mean wind speed · k: shape factor · Γ : gamma function

A2 · ANNUAL WIND ENERGY

$$E_{\text{wind}} = N \cdot P_{\text{rated}} \cdot 8.760 \cdot CF \cdot A$$

Availability is applied after the aerodynamic screening capacity factor.

A3 · HYDROGEN STOICHIOMETRY

$$m_{\text{H}_2}/m_{\text{MeOH}} = 3 \cdot M_{\text{H}_2}/M_{\text{CH}_3\text{OH}} = 188.75 \text{ kg/t}$$

Derived from $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$.

APPENDIX A • CONTINUED

Process and economics equations

A4 • CO₂ STOICHIOMETRY

$$m_{\text{CO}_2}/m_{\text{MeOH}} = M_{\text{CO}_2}/M_{\text{CH}_3\text{OH}} \approx 1.374 \text{ t/t}$$

CO₂ flow is derived from production; it is not set independently.

A5 • NET PRESENT VALUE

$$\text{NPV} = -\text{CAPEX}_{\text{net}} + \sum \text{FCF}_t / (1 + \text{WACC})^t$$

In the screening model, net CAPEX is placed at COD and operating cash flows are discounted over 25 years.

A6 • LEVELISED COST OF E-METHANOL

$$\text{LCOeM} = [\text{CAPEX}_{\text{net}} + \text{PV}(\text{OPEX}) + \text{PV}(\text{Decom})] / \text{PV}(m_{\text{MeOH}})$$

All economic and production terms use the same physical and time basis.

Reproducibility criterion

The appendix equations are the same conceptual rules implemented in the Python engine and web application. The aim is not to replace high-fidelity models, but to prevent an economic or production figure from becoming disconnected from its physical balance.

APPENDIX B

Correction register against the original thesis

ID	Correction	Treatment in this edition
C-01	51,058 t/y production attributed to a 15 MW concept.	Separate 15 MW unit / 150 MW hub and energy-limited calculation.
C-02	Strict carbon-neutrality claim.	Replaced by LCA/well-to-wake and RFNBO requirements.
C-03	Overly conclusive regulatory compliance/validation.	DNV/IEC treated as design basis; no class approval claimed.
C-04	40% Innovation Fund support treated as a near-fixed enabler.	Support made editable; call rules must be verified.
C-05	CfD presented as a necessary/guaranteed assumption.	Offtake and support treated as contractual scenarios, not certainties.
C-06	Aerodynamic, ventilation and ATEX claims unsupported by CFD/HAZOP.	Removed from conclusions; converted into future engineering gates.
C-07	Quantification of marine DAC degradation and prefiltration efficiency without project validation.	Not used as a performance basis; retained as a marine-integration risk.
C-08	NPV/IRR derived from financial bases that were not always consistent.	Single DCF model tied to the physical result.
C-09	Site suitability claimed without complete official GIS.	Site = screening only; permitting and conflicts remain open.

APPENDIX C

Primary-source and link register

- R01 · Gaertner, E. et al. (2020). IEA Wind TCP Task 37: Definition of the IEA 15-Megawatt Offshore Reference Wind Turbine. NREL. DOI 10.2172/1603478.
<https://doi.org/10.2172/1603478>
- R02 · U.S. Department of Energy. Technical Targets for Proton Exchange Membrane Electrolysis. Live PEM system-target page; accessed 13 Sep 2026.
<https://www.energy.gov/cmei/fuels/technical-targets-proton-exchange-membrane-electrolysis>
- R03 · IRENA & Methanol Institute (2021). Innovation Outlook: Renewable Methanol.
<https://www.irena.org/publications/2021/Jan/Innovation-Outlook-Renewable-Methanol>
- R04 · Copernicus Climate Change Service / ECMWF. ERA5 hourly data on single levels from 1940 to present. DOI 10.24381/cds.adbb2d47.
<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>
- R05 · EMODnet Bathymetry. Harmonised European DTM; 2024 DTM and OGC services.
<https://emodnet.ec.europa.eu/en/bathymetry>
- R06 · EMODnet Human Activities. EU Vessel Density Map - Detailed Method.
<https://emodnet.ec.europa.eu/en/human-activities-eu-vessel-density-map-detailed-method>
- R07 · MITECO. Spanish Maritime Spatial Plans (POEM).
<https://www.miteco.gob.es/es/costas/temas/proteccion-medio-marino/ordenacion-del-espacio-maritimo.html>
- R08 · Puertos del Estado. Oceanography / Portus: buoys, waves, sea level, currents and marine climate.
<https://www.puertos.es/en/services/oceanography>
- R09 · NREL. 2024 Electricity Annual Technology Baseline - Offshore Wind.
https://atb.nrel.gov/electricity/2024/offshore_wind
- R10 · International Energy Agency (2022). Direct Air Capture 2022.
<https://www.iea.org/reports/direct-air-capture-2022>
- R11 · IMO (2023). Strategy on Reduction of GHG Emissions from Ships, MEPC 80.
<https://www.imo.org/en/ourwork/environment/pages/2023-imo-strategy-on-reduction-of-ghg-emissions-from-ships.aspx>
- R12 · Regulation (EU) 2023/1805 - FuelEU Maritime.
<https://eur-lex.europa.eu/eli/reg/2023/1805/oj/eng>
- R13 · Commission Delegated Regulation (EU) 2023/1184 - renewable electricity rules for RFNBOs: additionality and temporal/geographical correlation.
https://eur-lex.europa.eu/eli/reg_del/2023/1184/oj/eng
- R14 · European Commission - Innovation Fund: programme, calls and competitive mechanisms.
https://climate.ec.europa.eu/eu-action/eu-funding-climate-action/innovation-fund_en
- R15 · European Commission - IF24 Call: call structure and results.
https://climate.ec.europa.eu/areas-action/eu-funding-climate-action/innovation-fund/calls-proposals/if24-call_en
- R16 · DNV (2021). DNV-ST-0119 Floating wind turbine structures - floating-stability update.
<https://www.dnv.com/news/2021/dnv-updates-standard-for-floating-wind-turbine-structures-to-allow-for-cost-reduction-and-optimization-205777/>
- R17 · Cui, X.; Kær, S.K.; Nielsen, M.P. (2022). Energy analysis and surrogate modeling for green methanol production under dynamic operating conditions. Fuel 307, 121924.
<https://doi.org/10.1016/j.fuel.2021.121924>
- R18 · Original BSc thesis by Aarón Expósito Monar (2026), Universidad de Cantabria. 171-page source document. Academic source document.
- R19 · MethaNor - interactive web application.
<https://aaronexposito0-creator.github.io/methanor-offshore-emethanol-digital-twin/>
- R20 · MethaNor - repository, engine, data, methodology and tests.
<https://github.com/aaronexposito0-creator/methanor-offshore-emethanol-digital-twin>

CLOSING

MethaNor: a design that can be challenged, recalculated and improved

A conceptual project is more useful when it states clearly what it knows, what it assumes and what it still has to prove. That is MethaNor’s role: turning an ambitious academic idea into an engineering-and-data architecture that can evolve in an auditable way toward pre-FEED, applied research or a demonstration proposal.

		
Interactive model	Code and evidence	LinkedIn

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